PROJECT WORK · KICK-OFF

PW: Sector scouting with patent data for an investment bank

Becoming a *patent analyst*

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Tuesday 6 October 2026

Politecnico di Torino

Project Work calendar

TUE 06 OCT	**Kick-off** 17:30–19:00 · Room R3
TUE 13 OCT	**Data collection: the TIP platform** 17:30–19:00 · Room R3
WED 21 OCT	**Landscaping tutorial 1: patent attributes** 13:00–14:30 · Room R3
MON 26 OCT	**Landscaping tutorial 2: patent text** 13:00–14:30 · Room 4P
MON 02 NOV	**Tutorial: knowledge spillovers** 13:00–14:30 · Room 4P
MON 09 NOV	**Tutorial: patent valuation, designing an algorithm** 13:00–14:30 · Room 4P

9 hours

of Project Work in class
6 sessions × 90 min

Companion lectures (+6 h)

20 Oct Patent intelligence and landscaping
28 Oct Knowledge spillovers
04 Nov Patent valuation
16 Dec Integrating strategy in landscape

Plus 25 h of tutor time (Dr. Giacomo Masali), booked by the teams

Why innovation data matters for a data scientist

The largest open dataset on technology

More than 150 million patent documents, plus scientific publications: structured metadata, citation networks, classification codes and full text. Ideal ground for NLP, network science and ML.

Data that drives real decisions

Investment banks, VCs, corporate strategy teams and policy makers use patent analytics to screen M&A targets, scout technologies and value intangible assets.

Hard, messy, real-world data

Name disambiguation, multiple jurisdictions, citation truncation, publication lags: exactly the problems data scientists are hired to solve.

Theory turns numbers into insight

A model without domain knowledge produces scores. Understanding appropriability, trajectories and dominant designs makes those scores defensible.

Patents (and publications) are the digital footprint of technological change. Learning to read them is a rare and valuable skill.

PW: sector scouting for an investment bank

You are consultants at an investment bank.

Your team is assigned a **technology sector**.
One of the bank's **clients**, an established company looking for growth, wants a clear picture of **how this sector is doing, overall and in its specific segments, and of the emerging players worth approaching as acquisition targets or partners.**

Your evidence: patent data

They show where firms invest in R&D years before products and revenues appear.

THE CLIENT'S THREE QUESTIONS

1. **How is the sector doing overall?**
2. **Which segments are growing, and which are fading?**
3. **Who are the emerging players to buy or partner with?**

The sector pool

- EV charging systems
- Quantum computing
- Humanoid robots
- Bioplastics
- Utility-scale energy storage
- Earth observation
- Smart glasses
- Precision agriculture
- Carbon capture
- Space debris removal
- Flying drones
- Underwater drones
- Surface drones
- Wave energy

Bank's clients

EV charging systems
Toyota

Quantum computing
Dell

Humanoid robots
Electrolux

Bioplastics
Johnson&Johnson

Utility-scale energy storage
Enel

Earth observation
CNH

Smart glasses
Amplifon

Precision agriculture
John Deere

Carbon capture
Holcim Cement

Space debris removal
SpaceX

Flying drones
Leonardo

Underwater drones
Fincantieri

Surface drones
STX France

Wave energy
Saipem

Expected output for the course: three deliverables



Presentation deck

A dense set of slides + technical appendix

- Sector landscape: technologies, volumes, trends, classes
- Competitive landscape and PEST analysis
- Present vs future scoring and Magic Quadrant
- Recommendation: target or partner, and why
- Technical appendix with all details that are not in the slides



Code

GITHUB REPOSITORY
(each team will be invited to join its repo)

- The full pipeline, versioned on GitHub
- Data collection, cleaning, indicators and scoring algorithm
- Reproducible: README, requirements,, run instructions



Pitch

10 MINUTES PER TEAM

- **Video:** mandatory for every team
- Format and deadlines explained in a later session

Presentation deck

1 General description

- Which technologies define the field?
- Patent-heavy or secrecy-reliant?
- Volumes, filing trends, CPC classes

2 Competitive landscape

- How concentrated is the field?
- Screen the players worth scoring
- PEST analysis

3 Scoring & Magic Quadrant

- **Present score**: influence, volume, success, globalization
- **Future score**: your own algorithm
- Map the actors on the quadrant

4 Strategic analysis

- Pick the most promising company
- Interpret it with theory
- Recommend: acquisition target or partner

Presentation Deck Step 0 · Data collection from TIP

Dedicated tutorial on data collection from TIP: **Tuesday 13 October**

Define the sector specific query

- **Start from CPC codes** that describe the technology (check the CPC scheme definitions)
- **Add keywords** in titles and abstracts to catch what the classes miss
- **Validate:** hand-label about 100 random families or find your own way to validate the query (for instance, look into the literature)

Extract (TIP · PATSTAT)

Most important tables:

- **tls201_appln:** applications, dates, grant flag, DOCDB family id
- **tls224_appln_cpc:** technology classes
- **tls206_person + tls207_pers_appln:** applicants and their countries
- **tls212_citation:** backward and forward citations
- **tls202 / tls203:** titles and abstracts for NLP

More tables will be presented during the tutorials

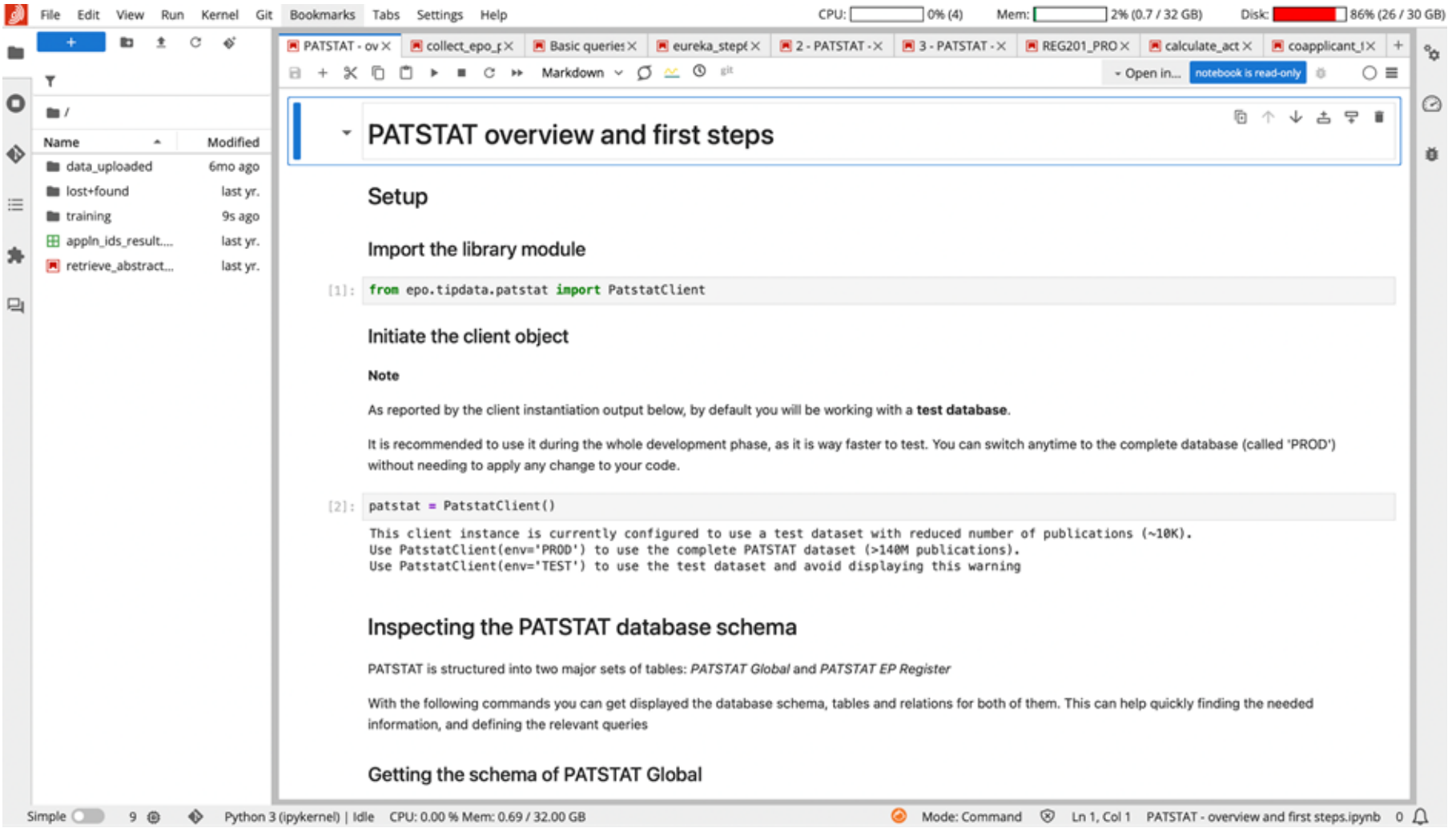
Clean

- **Unit of analysis:** the DOCDB family, so each invention is counted once
- **Time:** earliest priority year
- **Applicants:** harmonise names (PATSTAT PSN / HAN fields, plus your own rules for subsidiaries)
- **Truncation:** the last ~3 years are incomplete because of the 18-month publication lag: flag them, do not read them as a decline

Very important: register (make your own account) on TIP: https://tip.epo.org/hub/login?next=%2Fhub%2F
If you don't register on TIP you will not be able to work
Useful resources: CPC (technological codes) dictionary → https://www.cooperativepatentclassification.org/home

Presentation Deck Step 0 · The TIP platform

EPO Technology Intelligence Platform: where you query PATSTAT and build your dataset



Presentation Deck Step 1 · General description of the sector

What does this field look like, and how does it use IP?

Questions to answer

- Which **technologies** make up the field?
- How **big is it**, and is it growing, flat or declining?
- Which **technology classes dominate**, and how is the mix shifting?
- Is the field patent-heavy or does it rely on secrecy?

How

- **Volume:** families per priority year, growth rate over the window
- **Technology mix:** share of families by CPC subclass, and its change over time
- **Geography:** filing offices and applicant countries
- **Appropriability:** compare the sector with other evidence on patents vs secrecy (you can look at academic papers on Google scholar)

In the deck (suggestions)

- **Figure:** families over time, with the truncated years shaded
- **Figure:** CPC subclass mix over time
- **Table:** top CPC subclasses, with their official definitions
- **One argued paragraph:** how firms in this sector protect their inventions

Course link Types of IPR and appropriability (5 Oct) · Patent intelligence and landscaping (20 Oct) · Technological trajectories (30 Sep)

Presentation Deck Step 2 · Competitive landscape

Who is competing, how concentrated the field is, where the action is, and which players deserve to be scored.

Questions to answer

- How **concentrated** is the field, and is it consolidating?
- **Who are the incumbents, and who are the new entrants**?
- Which **subfields** are growing fastest?
- Which political, economic, social and technological forces shape it?

How

- **Concentration:** Herfindahl–Hirschman Index (HHI) (or others you find in the literature)
- **Entry:** new applicants per year (they never patented before) and their share of families + rate of disappearance
- **Subfields:** growth of CPC groups, or topics from abstracts (embeddings + clustering, topic models)
- **PEST analysis**

In the deck (suggestions)

- **Table:** top applicants, with type (firm, start-up, university) and country
- **Figure:** concentration and entry over time
- **Figure:** subfield growth map
- **Table:** PEST analysis (mandatory)

Deck Step 3 · Score the present value of top firms

1) **Pick the most interesting companies you find (you select how many, according to the story you think is worth telling)**
2) **You create a score for their value at present time using the four main areas (up to you to include the single indicators)**

INDICATOR	WHAT IT CAPTURES	HOW TO COMPUTE IT (PER FIRM- SUGGESTED, FEEL FREE TO ADAPT!)	KEY SOURCE
Influence	Technological impact	Forward citations received within 5 years, divided by the mean of the same CPC subclass × priority-year cohort; averaged over the firm's families	Hall, Jaffe & Trajtenberg (2001, 2005)
Volume	Weight in the sector	Firm's DOCDB families ÷ all sector families in the considered time window	Ernst (1998)
Success	Quality as judged by examiners (or commercial success)	Granted applications ÷ filed applications, on filings old enough to have been decided	Guellec & van Pottelsberghe (2000)
Globalization	Breadth of market protection	Mean number of distinct jurisdictions per family	Harhoff, Scherer & Vopel (2003)

Deck Step 4 · Score the future value of top firms

This is **your own** algorithm. Grants and citations do not exist yet for recent filings: score the last 1–2 years of published applications with forward-looking signals.

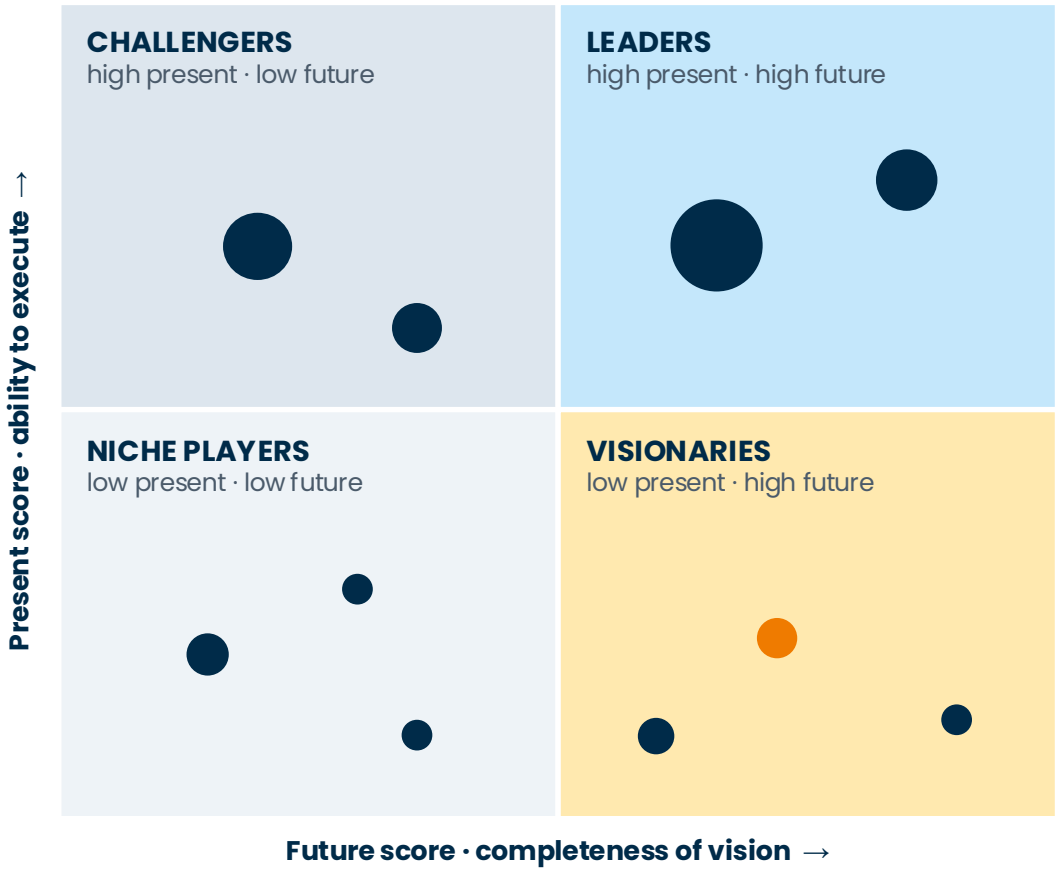
POSSIBLE INGREDIENTS · COMBINE, IMPROVE, OR REPLACE THEM

Originality	$1 - \Sigma s^2$ over the CPC classes of the patents it cites **Trajtenberg, Henderson & Jaffe (1997)**
Generality	Same formula over the patents that cite it; needs forward citations, so use with care on recent filings **Trajtenberg et al. (1997); Hall et al. (2001)**
Novel CPC recombination	Pairs CPC codes that have never co-occurred before **Fleming (2001); Verhoeven, Bakker & Veugelers (2016)**
Early filing in hot subfields	Enters a fast-growing CPC group ahead of the sector's curve **Course construct: tutorial of 9 Nov**
Text-based novelty	Low similarity to prior art, high similarity to what comes after (NLP) **Kelly et al. (2021); Arts, Hou & Gomez (2021)**

EVEN BETTER: PROVE IT WORKS (SUGGESTIONS)

1. Freeze time at T (e.g. 2015) and compute your score using only data available at T.
2. Measure what actually happened after T: 5-year forward citations, family growth, grants, the firm's later activity.
3. Compare with baselines from the literature
4. **Beat the baselines → a very high mark on valuation methodology, and a candidate for publication.**

Deck Step 5 · Build the Magic Quadrant



Illustrative: bubble size = number of families; orange = recommended firm

WHERE IT COMES FROM

Gartner's Magic Quadrant is a market-research method that places vendors on two axes, ability to execute (present score) and completeness of vision (future score), splitting them into Leaders, Challengers, Visionaries and Niche Players (Gartner, Magic Quadrant research methodology).

We replace analyst judgement with patent indicators, in the tradition of patent portfolio matrices (Brockhoff, 1992; Ernst, 1998).

Deck Step 6 · Strategic analysis and recommendation

Pick **the most promising company** on the quadrant, understand why it is strong, and tell the client what to do.

Choose

- Justify the pick from its position on the quadrant, not from its fame
- A Visionary may be an acquisition target; a Leader or Challenger may be a partner: argue your case

Explain: core competences

- **Access to many markets:** breadth of CPC classes and application domains
- **Customer value:** link patented technologies to product features and segments
- **Hard to imitate:** specialisation (revealed technological advantage, Patel & Pavitt, 1997), citation centrality, persistence over time

Recommend

- **Acquire or partner?** Weigh fit with the client's competences and complementary assets (Teece, 1986)
- **Three arguments** for the recommendation, backed by your data
- **Two risks:** e.g. litigation, freedom to operate, dependency, valuation
- Close the loop: answer the client's three questions

The deck · format and rules

20
slides in the main deck

+ Appendix
all the technical details

11 pt
technical appendix format

≤ 10
figures and tables

TECHNICAL APPENDIX

- Everything behind the slides: full explanations, formulas, references to scientific articles, and the full data construction pipeline. There are no specific requirements for the technical appendix as long as it justifies and contains everything you use.

GOOD PRACTICE WE WILL CHECK

- **Title slide:** team number, sector, members with student IDs, repository URL and commit hash
- **Citations:** consistent reference style
- **Figures:** clear, catchy, well-executed
- **Statements:** who did what (contribution statement) and how generative AI was used

The code · a reproducible GitHub repository

SUGGESTED STRUCTURE

team-XX-sector/
- **README.md** question, how to run, results
- **requirements.txt** pinned versions
- **config/sector.yaml**
- **src/**
  - **01_extract.py** SQL queries run on TIP
  - **02_clean.py** families, applicants, dates
  - **03_landscape.py** Steps 1–2
  - **04_present_score.py** Step 3, y-axis
  - **05_future_score.py** your algorithm
  - **06_quadrant.py** Steps 3–4 figures
- **notebooks/** exploration only
- **tests/** unit tests for indicators
- **outputs/** figures/ and tables/

REQUIREMENTS

- **Config, not hard-coding**
- **Data:** do not commit bulk PATSTAT extracts; commit the queries
- **Shared** with the teaching team (public, or private with access granted)

You can all commit or just whoever wants from the team!

The pitch · 10 minutes for the client's board

Your audience is not technical: they want the answer, the evidence and the risk. **No code on the slides**.

0–1 min
Hook
The sector and the client's question

1–3 min
Landscape
Size, growth, technologies

3–5 min
Competition
Who plays, where the action is

5–8 min
Quadrant
How you scored, what it shows

8–10 min
Recommendation
The company, acquire or partner, risks

Video · mandatory

- Every team submits one
- Hard limit 10:00
- Slides with voice-over; cameras optional
- Every member speaks

What makes a good pitch

- One message per slide
- Every claim traceable to the deck
- The quadrant is your centrepiece
- End on a decision, not a summary

Submission checklist

One package per team, uploaded by Sunday 20 December 2026 (not after). The upload channel will be announced in a later session.

Presentation deck	TeamXX_Sector_deck.pdf (slides + technical appendix)
Repository	URL containing full pipeline as required
Pitch video	TeamXX_Sector_pitch.mp4, ≤ 10:00

Submission deadline

DECEMBER 2026

20

Sunday

Submit by this date.
Not after.

One deadline for all three deliverables.

Upload the **presentation deck** (PDF, with technical appendix), the **repository** link with the commit hash of your v1.0-submission tag, the **pitch video** and the pitch slides by 20 December 2026. **Submissions after this date will not be accepted.**



Beyond the grade: where your work *could* go

Deck › a portfolio piece

An investment-grade sector study to show at interviews for consulting, IP analytics, VC and corporate strategy roles.

Code › a public, reproducible repository

Proof that you can build an end-to-end pipeline on a large, real-world dataset: a stronger signal than any certificate.

Pitch › talking to decision makers

Explaining a technical analysis to a non-technical client in 10 minutes is a core skill for any data scientist.

THE FUTURE-SCORE ALGORITHM

An open research problem

Predicting which patents and firms will matter tomorrow is still open in the literature: existing forward-looking indicators (originality, generality, novel recombination, text-based novelty) have well-known limits.

Improve on the literature → a very high mark on valuation methodology

Scientifically original projects can be developed further, with our support, into a scientific publication or a conference presentation.

PW wrap-up: from query to recommendation

0 STEP 0
Build the dataset
From TIP platform

CODE

1 STEP 1
Describe the field
Volumes, trends, technology classes, patents vs secrecy

CODE DECK

2 STEP 2
Map competition
Concentration, entrants, subfields, PEST

CODE DECK

3 STEP 3
Score the present
Influence, volume, success, globalization for the top firms

CODE DECK

4 STEP 4
Score the future
Your own algorithm on recent filings

CODE DECK

5 STEP 5
Magic Quadrant
Place the scored firms on the two axes

CODE DECK

6 STEP 6
Recommend
Core competences, acquire or partner, risks

DECK

PITCH
Tell the story
10-minute video for the client's board

PITCH

Deadline **Sunday 20 December 2026: deck, code and pitch video are all submitted by this date, not after.**

Evaluation criteria

20%	**Data & code quality** Collection, cleanliness, reproducibility		3 pts
25%	**Landscape, trajectory & market-opportunity analysis** Depth of the landscape		3.75 pts
20%	**Valuation methodology** Soundness and originality of the algorithm		3 pts
20%	**Integration with lecture topics** Entanglement with the course		3 pts
15%	**Presentation quality** Clarity and a defensible, reasoned narrative		2.25 pts



15

POINTS MAX

Top marks on valuation methodology go to algorithms that improve on the literature.

Who supports you



Prof. Grazia Sveva Ascione

LECTURER · PROJECT WORK LEAD

- Project brief, methodology and tutorials
- Final evaluation of the project work

grazia.ascione@polito.it



Dr. Giacomo Masali

TUTOR · 25 HOURS FOR THE TEAMS

- Schedule meetings with him for your questions on data, code and methodology
- About 35–40 min per team over the whole project: book early, come with specific questions

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Fatima Syeda Wasti

STUDENT TUTOR

- Team formation and group assignment
- Administrative matters and logistics

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Useful references

MAGIC QUADRANT AND PATENT PORTFOLIOS

Gartner. (n.d.). Magic Quadrant research methodology. Gartner. https://www.gartner.com/en/research/methodologies/magic-quadrants-research

Brockhoff, K. K. (1992). Instruments for patent data analyses in business firms. Technovation, 12(1), 41–59. https://doi.org/10.1016/0166-4972(92)90031-C

Ernst, H. (1998). Patent portfolios for strategic R&D planning. Journal of Engineering and Technology Management, 15(4), 279–308. https://doi.org/10.1016/S0923-4748(98)00018-6

Squicciarini, M., Dernis, H., & Criscuolo, C. (2013). Measuring patent quality: Indicators of technological and economic value (OECD Science, Technology and Industry Working Papers No. 2013/03). OECD Publishing. https://doi.org/10.1787/5k4522wkw1r8-en

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Useful references

FUTURE-SCORE INDICATORS

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STRATEGY AND APPROPRIABILITY

Cohen, W. M., Nelson, R. R., & Walsh, J. P. (2000). Protecting their intellectual assets: Appropriability conditions and why U.S. manufacturing firms patent (or not) (NBER Working Paper No. 7552). National Bureau of Economic Research. doi: 10.3386/w7552

Arundel, A. (2001). The relative effectiveness of patents and secrecy for appropriation. Research Policy, 30(4), 611–624. doi: 10.1016/S0048-7333(00)00100-1

Teece, D. J. (1986). Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. Research Policy, 15(6), 285–305. doi: 10.1016/0048-7333(86)90027-2

Utterback, J. M., & Abernathy, W. J. (1975). A dynamic model of process and product innovation. Omega, 3(6), 639–656. doi: 10.1016/0305-0483(75)90068-7

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Patel, P., & Pavitt, K. (1997). The technological competencies of the world's largest firms: Complex and path-dependent, but not much variety. Research Policy, 26(2), 141–156.

Questions?

Next session

Data collection: the TIP platform · Tuesday 13 October, 17:30–19:00 · Room R3



Politecnico di Torino